The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “internal energy” and “work” to explain your answer. Sentence starters are optional.

internal energy: energy stored in a system’s microscopic motion/interactions

work: energy transferred by a force along a displacement

surroundings: everything outside the chosen system

01 Warm-up: bring back an earlier idea

A **3 kg** sample completely changes phase with latent heat **500 J/kg**. **Find the energy magnitude required**.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives **90 J** of heat and does **30 J** of work on its surroundings. With ΔU=Q-W, internal energy increases by **60 J**.

Your turn: Draw labeled arrows crossing a system boundary and **mark the 60 J** stored increase.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

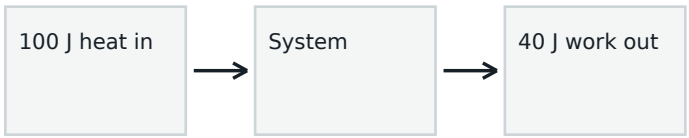
Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A system absorbs **100 J** of heat and does **40 J** of work. **Draw the energy arrows** and **find its internal-energy change**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A gas in a piston absorbs **80 J** of heat and does **20 J** of work by pushing outward.
1. **Label the input**, work output and stored-energy change.
2. Use $\Delta U = Q - W_{by}$ to find whether its internal energy increases or decreases.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why does the phrase “work is **60 J**” need a direction or sign convention?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Total kinetic energy changes from **24 J** to **20 J** in an isolated collision. Is it elastic? Where could the difference go?
Show the energy changes and transfers; **label units**.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Heat input **30 J**; work output **10 J**.
1. Find ΔU.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Give two different heat-input**/work-output pairs that increase internal energy by **40 J**.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3